

Working

Protocols

Organisation Structure, Roles & Operating Logic

A joint venture between Pantaleon and Findability Sciences

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1. Introduction

Stomata Labs is a joint venture between **Pantaleon**, a leading sugar mill group with deep operational expertise in sugarcane agriculture, industrial processing, and commercial logistics, and **Findability Sciences**, an applied AI company specialising in machine learning, data engineering, and software product development. The venture exists to build and commercialise a suite of analytics and AI products — the Stoma platform — that transform how sugar mills make decisions.

This document describes how Stomata Labs is organised to deliver on that mission. It explains the logic behind the organisational structure, defines every role, and describes how the different functions interact operationally. It is intended for all team members — technical and commercial — as well as the Steering Committee. Readers should finish this document with a clear understanding of who does what, why the structure is designed this way, and how individual contributions connect to the venture's commercial objectives.

A companion interactive tool — the Working Protocols HTML guide — provides a searchable, filterable reference for all 57 defined operating protocols. This document provides the reasoning and context that the interactive tool does not.

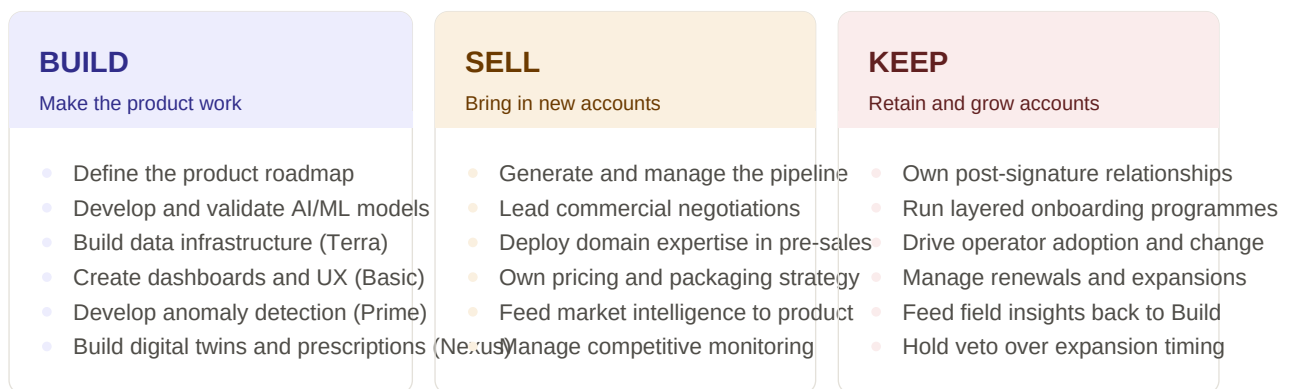
→ **Note for Steering Committee.** This document is the foundation for Stomata Labs' OKR process. Every role described here maps to a set of measurable objectives. Section 8 explains how to use this structure to set OKRs at the team and individual level.

2. Why this organisation

The structure of Stomata Labs was designed from first principles, starting with a simple question: *what does this venture need to do in order to succeed?* The answer has three non-negotiable components.

2.1 The Build / Sell / Keep framework

Any B2B software venture can fail in exactly three ways: it fails to build a product that works, it fails to sell that product to customers, or it fails to keep customers after they have signed. These are not sequential risks — they operate simultaneously from the first day. The organisational structure maps directly onto these three failure modes.



2.2 Why each pillar matters for Stomata

Build matters because the product is technically ambitious. The Stoma platform spans four layers of increasing complexity — from data infrastructure through descriptive analytics, predictive AI, and prescriptive optimisation. Building this requires a team that combines deep software engineering, industrial AI expertise, and domain knowledge of sugar mill processes. Without a strong Build pillar, the product will not be credible enough to sell or retain.

Sell matters because the market is specialised. The global sugar mill industry is a tight-knit community of technically sophisticated buyers. A standard enterprise software sales motion does not work here. Every conversation requires domain credibility — the ability to discuss POL optimisation, extraction efficiency, or crystallisation dynamics at the level of a process engineer. This is why the Sell pillar is built around Domain Solution Engineers who carry both technical and domain depth.

Keep matters because sugar mills are inherently conservative. Industrial operators do not change how they make decisions quickly. Signing a contract does not mean adoption. A mill that has installed the product but does not use it is a churn risk, not a success story. The Keep pillar exists to close the gap between deployment and genuine adoption — which requires a structured change management capability, not just account management.

■ The three pillars are not independent. A Build team that does not receive field feedback from Keep will build the wrong things. A Sell team that promises capabilities Build cannot deliver will create churned accounts for Keep to manage. The operating protocols in this document exist to ensure information flows correctly between all three pillars at all times.

3. Product architecture — the four Stoma layers

The Stoma platform is a progressive intelligence stack. Each layer answers a different operational question and builds on the layer below it. This architecture has direct implications for the organisation — different roles are responsible for different layers, and the commercial strategy is built around a land-and-expand motion that moves accounts progressively from Terra through to Nexus.

STOMA NEXUS

Prescriptive — What needs to be done?

Digital twins · world models · setpoint prescriptions

STOMA PRIME

Predictive — What will happen?

Anomaly detection · time/frequency domain · reconstruction error

STOMA BASIC

Descriptive — What is happening?

Dashboards · SPC charts · trend views · real-time alerts

STOMA TERRA

Integration — How does data get here?

IoT sensors · satellite · climate APIs · cloud data lake

STOMA TERRA

Terra is the foundational data infrastructure layer. It connects IoT sensors on the factory floor and in the field, ingests satellite data and climate API feeds, and consolidates everything into a unified cloud data lake. Terra is a prerequisite for every other layer. It is also the most complex implementation challenge at each new client site, because every mill has a different sensor infrastructure, connectivity environment, and IT maturity level. A third-party IoT integration platform reduces bespoke engineering, with the Data/Backend Engineer owning the platform configuration, data contracts, and quality SLAs.

STOMA BASIC

Basic is the descriptive analytics layer, answering: what has happened, and what is happening right now? The product delivers online dashboards showing time-series data, operational trends, and Statistical Process Control (SPC) charts. Basic is the recommended land product in a new mill account — it is the fastest to deploy, requires the least organisational change from operators, and produces visible value quickly. Basic builds the data trust and operator familiarity that Prime and Nexus depend on.

STOMA PRIME

Prime is the predictive analytics layer, answering: what will happen? Prime applies industrial AI methods — time and frequency domain analysis, autoencoders, PCA-based reconstruction error, and cross-variable correlation — to detect anomalies in vibration, temperature, and process signals before they become failures. Prime quality depends entirely on Terra data quality — an explicit SLA between the two layers must be defined and enforced before any Prime deployment. Prime is owned by the Signals ML Engineer, a specific industrial AI profile rather than a generalist data scientist.

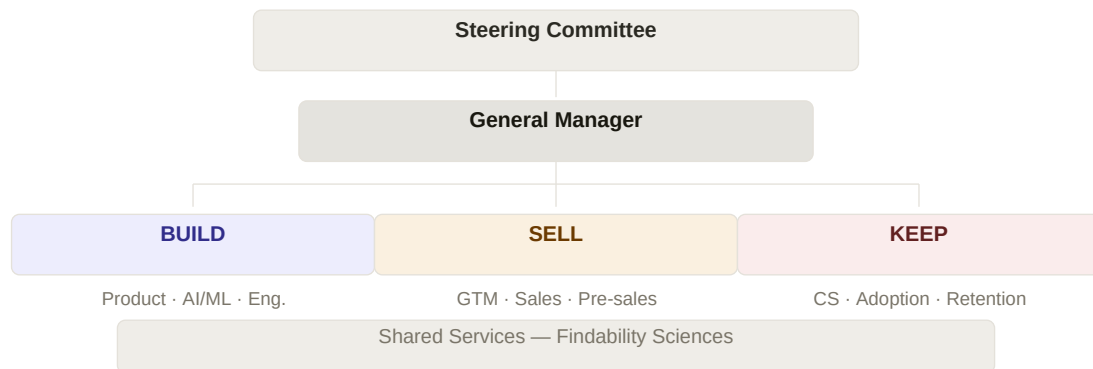
STOMA NEXUS

Nexus is the prescriptive layer, answering: what needs to be done? Nexus builds world models and digital twins of mill processes, then applies optimisation solvers to generate specific operating setpoint prescriptions for mill operators. This is at the frontier of commercially deployed industrial AI. Nexus requires deep domain validation — no prescription can reach an operator without confirmation from the Domain Data Scientist that it is physically plausible and operationally safe. Nexus is positioned as a co-development engagement with early accounts rather than a standard product delivery.

→ Commercial sequencing: land with Terra + Basic, establish data trust, then introduce Prime when sensor data coverage is good and the maintenance team is receptive. Nexus follows only when Prime adoption is solid and the process engineering team is ready to act on AI prescriptions. No layer should be sold before the previous one is genuinely adopted. The CS & Adoption Lead holds veto authority over expansion timing.

4. Organisational structure

The structure below shows how Stomata Labs is organised at launch — a deliberately lean design appropriate for early commercialisation. The three pillars sit at the same level under the General Manager, supported by Findability Sciences shared services. As the venture scales, the structure will evolve — specific triggers for role splits are identified in Section 5.



Stomata Labs organisational structure.

4.1 Governance

The **Steering Committee** — composed of the CEOs of Pantaleon and Findability Sciences — holds strategic authority. It meets monthly and is responsible for budget approvals, partner commitment decisions, market expansion choices, and disputes between the two partner organisations that cannot be resolved operationally. The Steering Committee is not a management layer and does not make day-to-day decisions.

The **General Manager** is the single accountable executive for the venture. This role holds P&L; responsibility, resolves cross-pillar conflicts, manages commercial vendor agreements (including Terra data providers), and ensures partner commitments are honoured. Without a clearly named GM, daily decisions escalate to a committee that meets monthly — a structural bottleneck that would slow the venture to a halt.

4.2 Build pillar

Build is responsible for everything the product is and everything it does — combining product management, AI/ML research, data engineering, and application development into a single accountable unit. The Head of Product & AI is the single person responsible for the product being technically sound, operationally credible, and safe to deploy at a sugar mill.

The Build team brings together engineering and ML talent from Findability Sciences with domain data science expertise from Pantaleon. The combination is essential — engineering talent without domain knowledge produces models that are technically correct but operationally useless. Domain knowledge

without engineering talent produces insights that cannot be productised.

4.3 Sell pillar

Sell owns the full commercial motion from pipeline generation through contract signature. The pillar is led by the GTM Lead and supported by three Domain Solution Engineers — one per domain (agriculture, industrial processing, logistics/commercial). Domain Solution Engineers are the venture's most distinctive commercial asset: technically credible engineers who can hold a peer-level conversation with a mill's process director, maintenance chief, or chief agronomist.

Domain Solution Engineers carry a dual accountability: they report to the GTM Lead for commercial activities, and maintain a formal input channel to the Head of Product & AI for product feedback. Their time is governed by a 70/30 protocol — 70% on Sell activities, 30% on product intelligence contribution — monitored monthly by the GM.

4.4 Keep pillar

Keep is responsible for everything that happens after a contract is signed. The CS & Adoption Lead owns account health, onboarding, change management, renewals, and the field feedback loop to Build. This role carries a layered mandate: the onboarding approach for Stoma Basic is fundamentally different from Stoma Nexus, where operators must trust and act on AI-generated prescriptions that affect process safety and professional accountability.

The Keep pillar is designed to scale. At six active mill accounts, the CS & Adoption Lead role splits into Account Manager (relationships and renewals) and Change Management & Training Specialist (onboarding and adoption programmes). This trigger is defined now so that it is planned proactively rather than discovered in crisis.

4.5 Shared services

Finance, legal, HR, DevOps/Platform Operations, QA, and security are provided by Findability Sciences as a shared service. Service levels and cost allocation must be explicitly defined in the joint venture agreement. One shared service warrants particular attention: **data isolation and security**. The multi-client Terra data lake architecture means that client data must be strictly isolated. This is a legal, ethical, and commercial requirement that must be defined before the first multi-client Terra deployment.

5. Roles and responsibilities

Each role in Stomata Labs has a defined purpose and a set of owned outcomes. Detailed role descriptions including success metrics are maintained in the role library alongside this document.

General Manager Governance

Single accountable executive. Owns P&L; resolves cross-pillar conflicts, manages vendor agreements, reports to Steering Committee. The operational authority that prevents the venture from being governed by committee.

Head of Product & AI Build pillar

Defines what gets built and to what standard across all four Stoma layers. Owns the product roadmap, AI/ML methodology, data governance, UX quality bar, and Terra data quality SLAs. Single person accountable for the product being technically sound and operationally credible.

Signals ML Engineer Build pillar

Designs and maintains Stoma Prime anomaly detection models using time/frequency domain methods, autoencoders, and reconstruction error approaches. A specific industrial AI profile. Owns Prime model quality and production monitoring.

Process Optim. ML Engineer Build pillar

Builds Stoma Nexus world models, digital twin representations, and optimisation solvers. Works intensively with the Domain Data Scientist to validate that every prescription is physically plausible and operationally safe. The most technically ambitious role in the venture.

Domain Data Scientist Build pillar

The bridge between domain expertise and model development. Validates every model assumption against operational mill reality. Critical for Nexus world model validation and agricultural remote sensing interpretation in Terra. Without this role, models will be technically correct but operationally wrong.

Data / Backend Engineer Build pillar

Owns the Terra data lake architecture, all ingestion pipelines, and backend API services powering Basic and Prime. Defines and enforces data quality SLAs. If Terra data quality is insufficient for Prime, this must be surfaced before a model is trained.

Front-end / App Engineer Build pillar

Builds the interfaces through which operators interact with Stoma outputs — from Basic dashboards to Nexus prescription screens. Requires UX sensibility for industrial environments. The quality of this layer determines whether operators trust and act on the product.

GTM Lead Sell pillar	Owns the full commercial motion from pipeline generation through contract signature. Leads pricing strategy, the layer-by-layer expansion commercial playbook, and coordinates Domain Solution Engineers by opportunity. Single person accountable for new revenue.
Domain Solution Eng. x3 Sell pillar	The technical credibility engine of the sales cycle. Three engineers — one per domain (agriculture, industrial, logistics) — lead pre-sales conversations at the level of mill engineers and process directors. Dual accountability to GTM Lead (commercial) and Head of Product & AI (product intelligence). Governed by 70/30 time split.
CS & Adoption Lead Keep pillar	Owns everything post-signature: onboarding, change management, renewals, field feedback, and expansion coordination. Layered onboarding mandate — a different programme per product layer. Holds veto authority over expansion timing. The role that determines whether deployment becomes adoption.

6. How the pillars work together

The three pillars are interdependent. Build produces what Sell promises and what Keep delivers. Sell generates the commercial intelligence that shapes Build's roadmap. Keep provides the field reality that validates and challenges both Build and Sell. The operating protocols that govern these interdependencies are described here at the functional level.

6.1 Management interactions

The General Manager operates on a structured rhythm. **Weekly:** pipeline review with GTM Lead; priority alignment with Head of Product & AI. **Bi-weekly:** account health review with CS & Adoption Lead. **Monthly:** shared services SLA review; DSE time allocation review; venture health report to Steering Committee. **Quarterly:** roadmap strategic review; pricing strategy review.

The GM's primary operational role is to resolve conflicts between pillars before they become blockers. The most common conflict at this stage will be between Build capacity and commercial commitments — the GM must ensure Sell is not promising capabilities Build cannot deliver, and Build is not building features disconnected from commercial reality.

6.2 Build interactions

The Build team operates on a sprint cadence, with the Head of Product & AI running weekly backlog grooming to align sprint goals with commercial priorities and field feedback. The most critical recurring Build interaction is the weekly session between the Process Optimisation ML Engineer and the Domain Data Scientist for Nexus model validation — this is the quality gate between a world model prescription and an operator acting on it.

Build receives inputs from two external sources: Domain Solution Engineers (bi-weekly debrief of what was learned in client conversations) and the CS & Adoption Lead (bi-weekly structured field feedback on what operators actually use, ignore, or misunderstand). Both channels must be protected — they are the mechanisms by which the product stays connected to operational reality.

6.3 GTM interactions

The GTM Lead runs a weekly pipeline review with the GM and bi-weekly alignment sessions with the Head of Product & AI. This alignment is commercially critical: it is where the GTM Lead learns what the product can and cannot do at each layer, which directly determines what Sell can honestly promise to prospects. Domain Solution Engineers debrief the GTM Lead after every client engagement — providing deal fit assessment, data availability findings, competitive intelligence, and a realistic view of layer scope achievability.

A formal DSE layer-readiness framework governs which Domain Solution Engineer can lead which conversation. Terra and Basic: all three DSEs independently. Prime: technical preparation required. Nexus: the Head of Product & AI must be present — no exceptions. On contract signature, the GTM Lead executes a formal written commercial handoff to the CS & Adoption Lead covering all commitments made during the sales process.

6.4 Customer Success interactions

The CS & Adoption Lead sees what operators actually do with the product — not what they say they do. The bi-weekly structured field feedback session with the Head of Product & AI is the primary mechanism for this intelligence to flow into the product roadmap.

Before any account is proposed for next-layer expansion, CS must confirm that the current layer is genuinely adopted — not just installed. This confirmation carries veto authority. For Nexus accounts, CS operates a 90-day intensive adoption programme with the Process Optimisation ML Engineer and the relevant Domain Solution Engineer as active ongoing participants. Nexus adoption requires operators to change their professional decision-making process — a change management challenge requiring sustained engagement from people with domain credibility.

7. Critical protocols

Of the 57 operating protocols defined for Stomata Labs, eight are designated as critical. These interactions must not be skipped, deprioritised, or treated as informal — the consequences range from failed deliveries to safety risks to churned accounts.

■ Process model validation with Domain DS

Scope: Build · Nexus

The only gate between a world model prescription and a mill operator acting on it. Must produce written validation outputs. If this becomes a casual check-in, an operationally unsafe or physically impossible setpoint can reach a client.

■ Domain knowledge debrief — field to product

Scope: Sell → Build · Cross

The mechanism by which product development stays connected to mill operational reality. If cancelled when calendars fill, model assumptions drift from the field and Prime/Nexus quality degrades invisibly.

■ Terra data quality gate enforcement

Scope: Build · Terra → Prime

Prime model performance depends entirely on Terra data quality. This gate must produce a written quality report before any Prime model is deployed. Skipping it under delivery pressure produces a degraded model that damages client trust.

■ Mandatory Nexus scoping co-presence

Scope: Sell · Nexus

No Nexus scoping conversation occurs without the Head of Product & AI present. One incorrectly scoped Nexus engagement can result in a failed delivery in a market where reputation travels fast.

■ CS layer adoption readiness confirmation

Scope: Keep → Sell · Cross

CS holds veto authority over expansion timing. Before the GTM Lead proposes next-layer expansion, CS must confirm the current layer is genuinely adopted. Expansion before adoption is the fastest path to churn.

■ Commercial handoff on contract signature

Scope: Sell → Keep · Cross

Everything promised during the sales process must transfer in writing to the person who will deliver it. A one-page written handoff document is mandatory — verbal handoffs produce broken onboarding expectations.

■ Nexus physical plausibility violation escalation

Scope: Build · Nexus

If the Domain Data Scientist identifies a world model prescription that violates physical process constraints, all work on that branch stops immediately. No timeline or commercial pressure overrides this.

■ Client data isolation architecture sign-off

Scope: Build · Terra

Before any new client data enters the Terra data lake, the Head of Product & AI must sign off on the isolation architecture. This is a legal, ethical, and commercial requirement.

8. Setting up for OKRs

The organisational structure described in this document provides the natural framework for Stomata Labs' OKR process. OKRs work best when anchored to clear accountability — a single owner per objective, measurable key results that reflect real operational outcomes, and a cadence that matches how the business actually moves.

OKR hierarchy

Stomata Labs OKRs should cascade across four levels.

Venture level	Set by the GM with Steering Committee. Covers: pipeline and revenue targets, product layer delivery milestones, customer retention and adoption rates. Reviewed quarterly.
Pillar level	One OKR set per pillar (Build, Sell, Keep), owned by the pillar lead. Each pillar objective should directly support a venture-level objective. Reviewed quarterly; key results tracked monthly.
Role level	Individual OKRs for each named role. Derived from pillar OKRs. Each person's key results should be measurable outputs of their defined responsibilities. Reviewed quarterly; checked monthly in 1:1s.
Protocol level	The 57 working protocols define operational excellence at the interaction level. Key results can reference protocol compliance — e.g. "DSE debrief delivered 100% of weeks" or "commercial handoff completed within 48h of every contract signature".

Suggested OKR anchors by pillar

The following are indicative objective areas for each pillar. Key results should be defined collaboratively with each pillar lead using specific, measurable targets appropriate to the venture's current stage.

Pillar	Objective areas	Indicative key result types
Build	Product layer delivery Model quality and reliability Data infrastructure UX and adoption	On-time layer delivery rate Model accuracy vs. agreed baseline Terra pipeline uptime SLA UI-related support tickets
Sell	Pipeline generation Deal conversion Layer expansion Market intelligence	Qualified pipeline value Win rate by layer Layer expansion per existing account DSE debrief quality and cadence

Keep	Account adoption Retention and renewal Field intelligence Nexus programme	Active usage rate per layer per account Net Revenue Retention Field feedback sessions delivered Nexus prescription acceptance rate
GM / Venture	Revenue Product credibility Team stability Partner governance	ARR vs. plan Accounts at full-layer adoption Employee retention Shared services SLA compliance

→ OKRs should be set after this document has been reviewed by all team members. The most effective first OKR cycle is one where each person's key results are directly traceable to a working protocol — linking daily operational behaviour to measured outcomes. The Steering Committee should review and approve the venture-level OKRs at the first quarterly review.